

## Mode of Speciation in Rose Finches

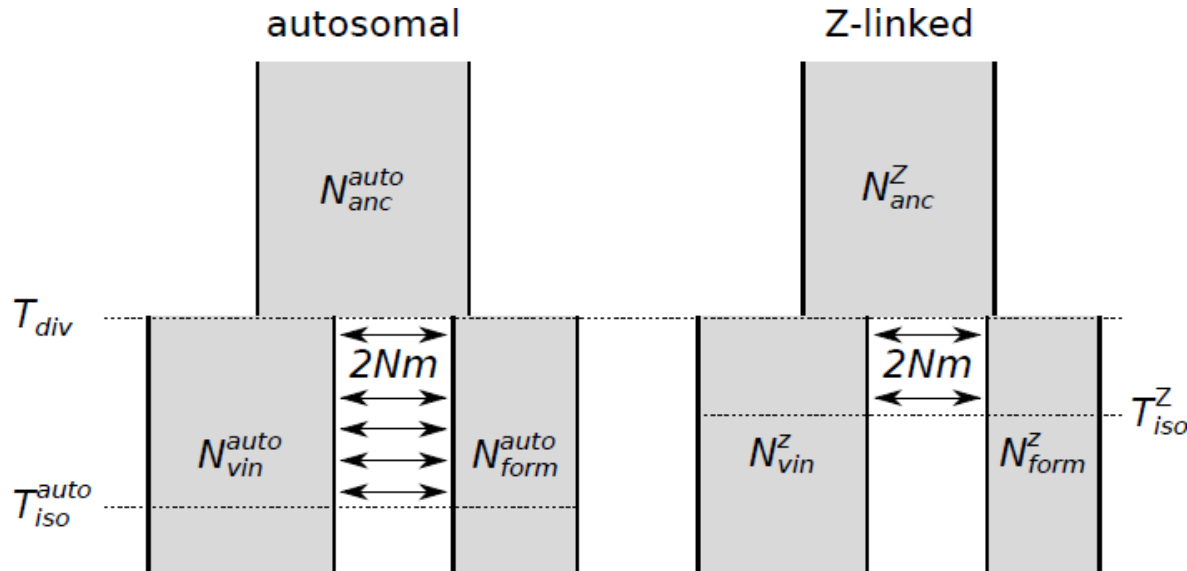
- In the classic view, **geographic isolation** was considered essential for speciation.
- However, recent evidence suggests that local adaptation and speciation may occur in the presence of **gene flow** if ecological selection is strong.
- In Birds, the **Z-chromosome** is known to play a vital role in speciation
  - **Haldanes Rule:** In hybrids, fitness is lower in the hemizygous sex (females)
  - Male **sexually selected traits and female preference** was mapped to the Z-chromosome in several species.
- **Prediction**  
If selection against hybrids is a driving force in speciation, gene flow will be interrupted earlier on the Z-chromosome than on autosomes.

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- Inferring isolation times for Z-linked and autosomal markers separately.



Shou-Hsien Li



*Carpodacus vinaceus* (Himalaya)



*Carpodacus formosa* (Taiwan)

## Two major difficulties

- For realistic evolutionary models, analytical solutions of the likelihood function are usually **very hard** and often **impossible** to obtain.
- We will use two tricks:
  - 1) Using **summary statistics** **S** instead of the full data **D**
    - The hope is that  $P(\mathbf{D} | \theta)$  is proportional to  $P(\mathbf{S} | \theta)$
  - 2) Using **simulations** to approximate the likelihood function  $P(\mathbf{S} | \theta)$

- Apply in a Bayesian setting: 
$$\underbrace{P(\theta | \mathbf{D})}_{\text{Posterior}} \quad \underbrace{P(\mathbf{D} | \theta)}_{\text{Likelihood}} \quad \underbrace{P(\theta)}_{\text{Prior}}$$



Approximate Bayesian Computation (ABC)

# Approximate Bayesian Computation ABC

defining statistics

Sequence Data



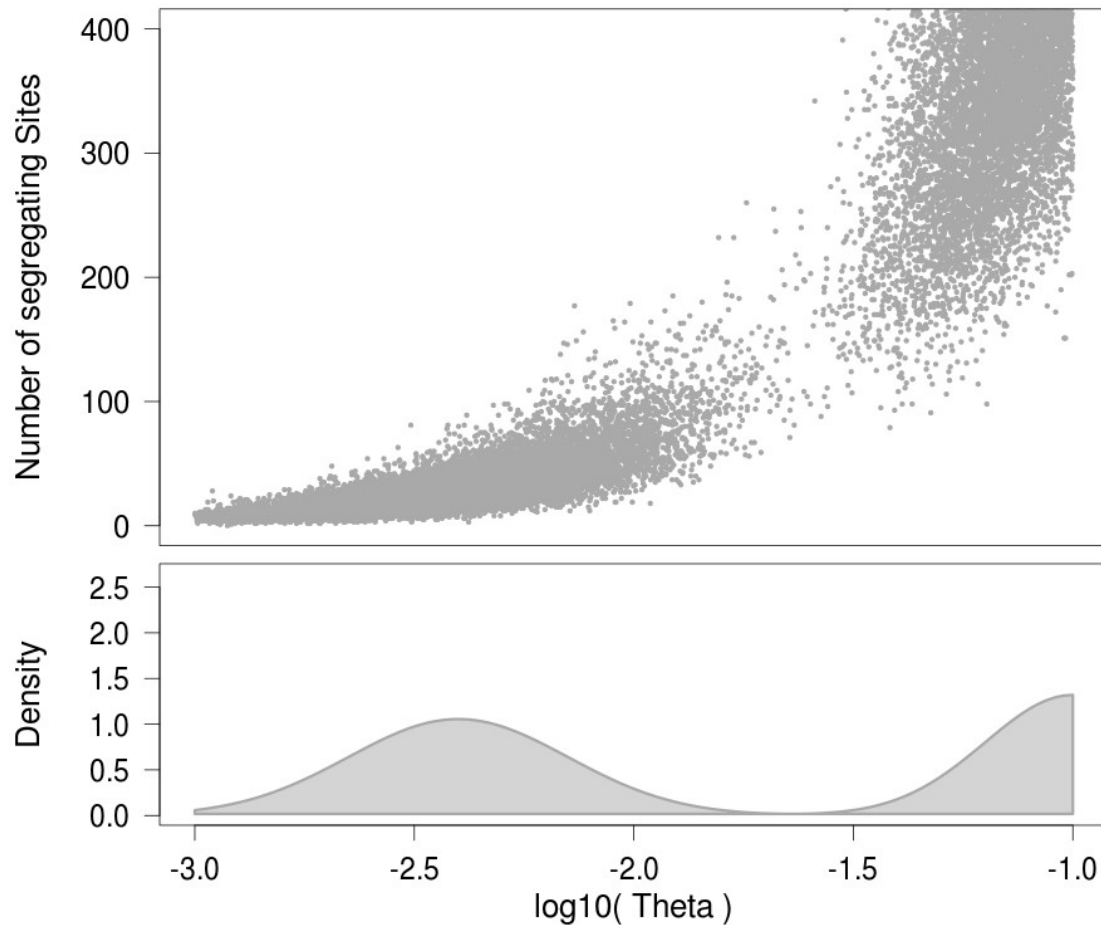
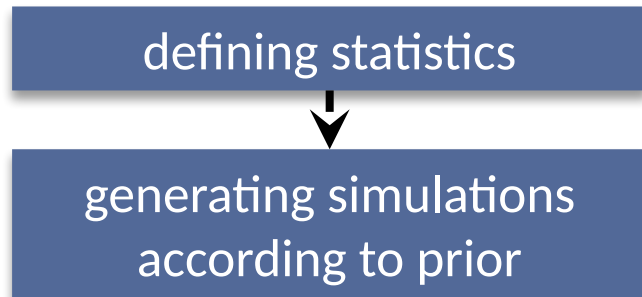
Data



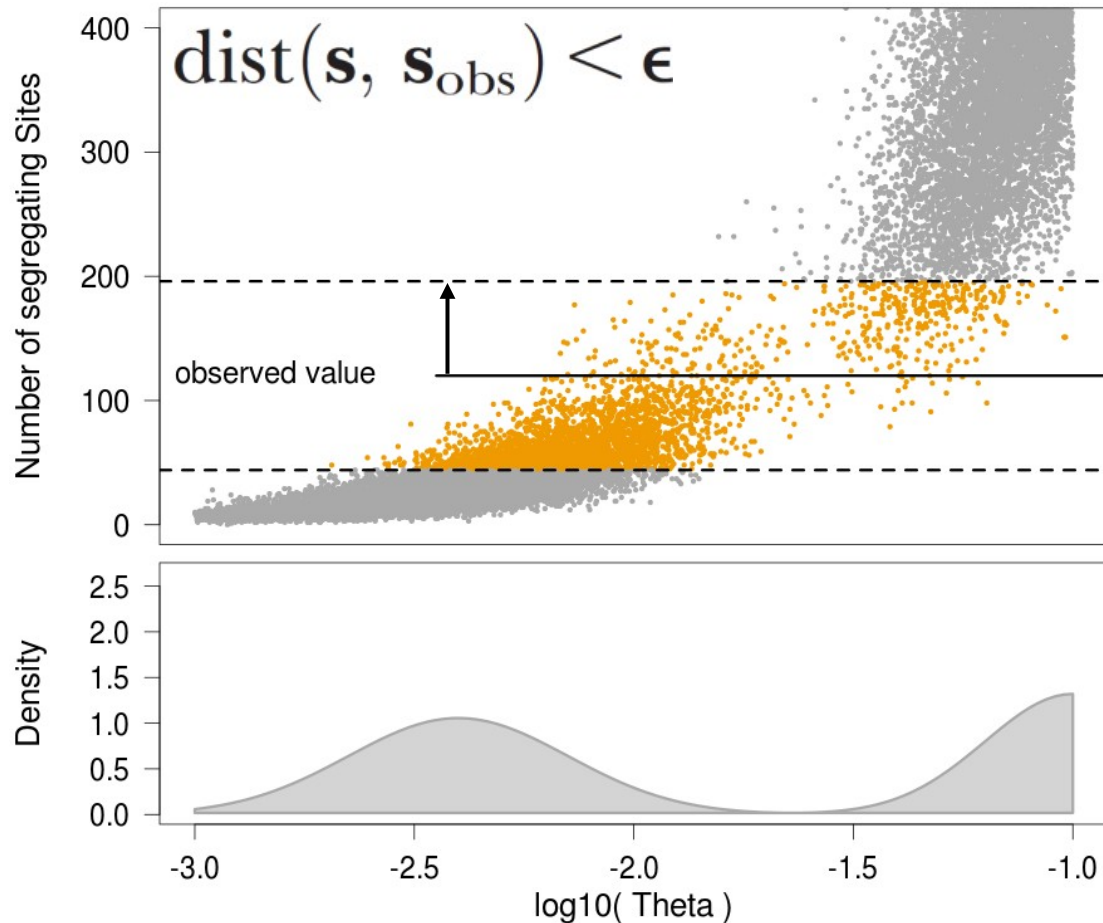
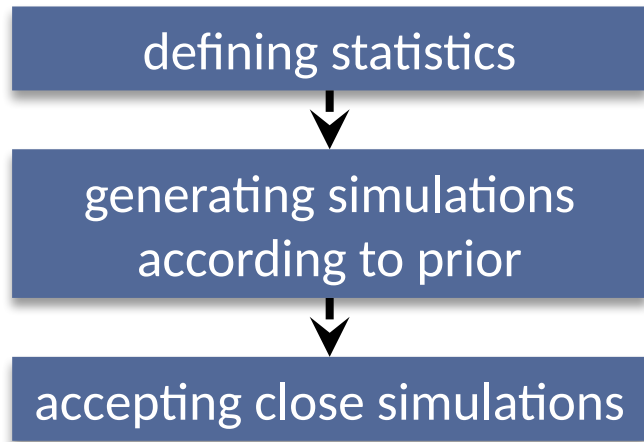
$S, F_{ST}, D, \dots$

Summary statistics

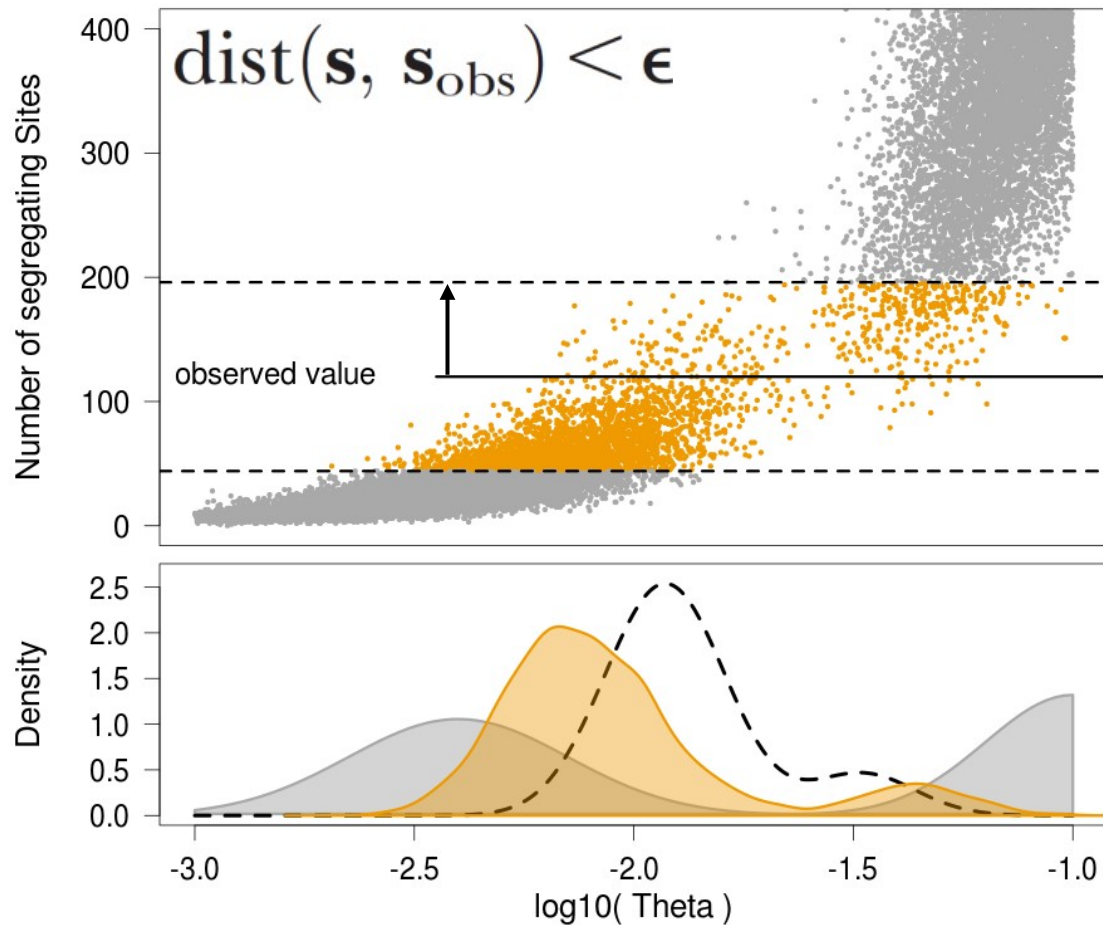
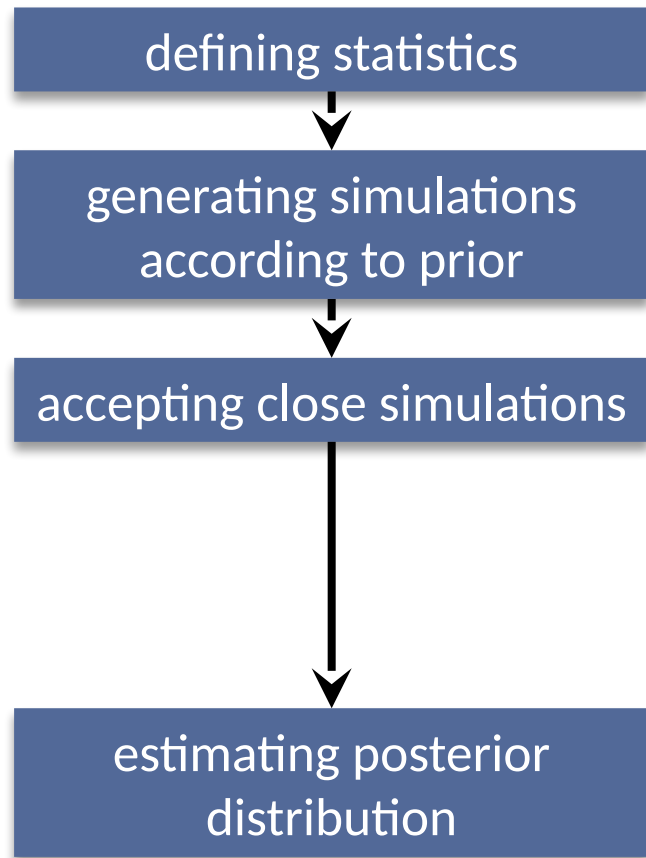
# Standard ABC Algorithm



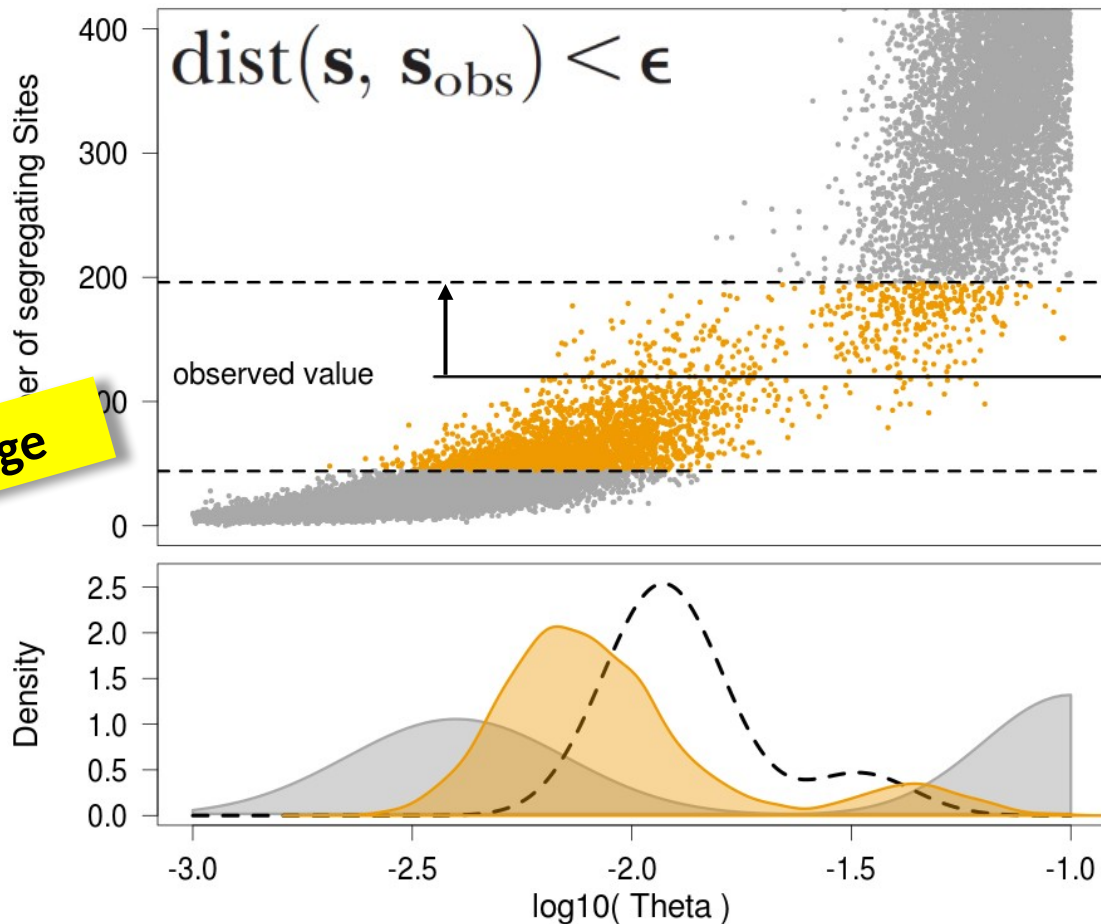
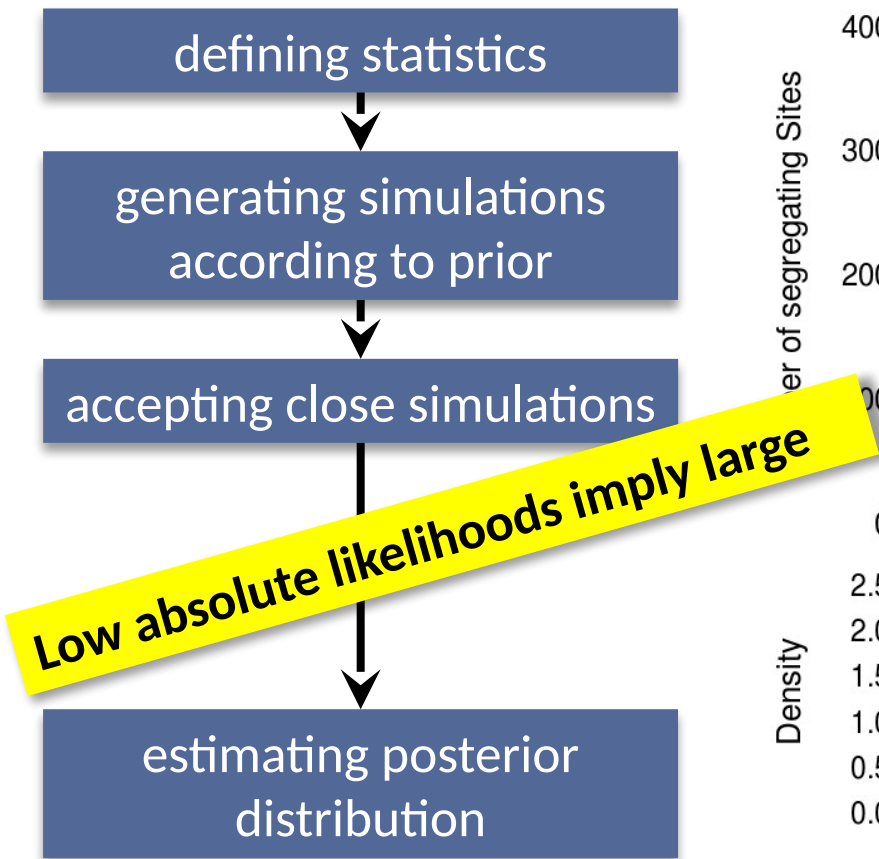
# Approximate Bayesian Computation ABC



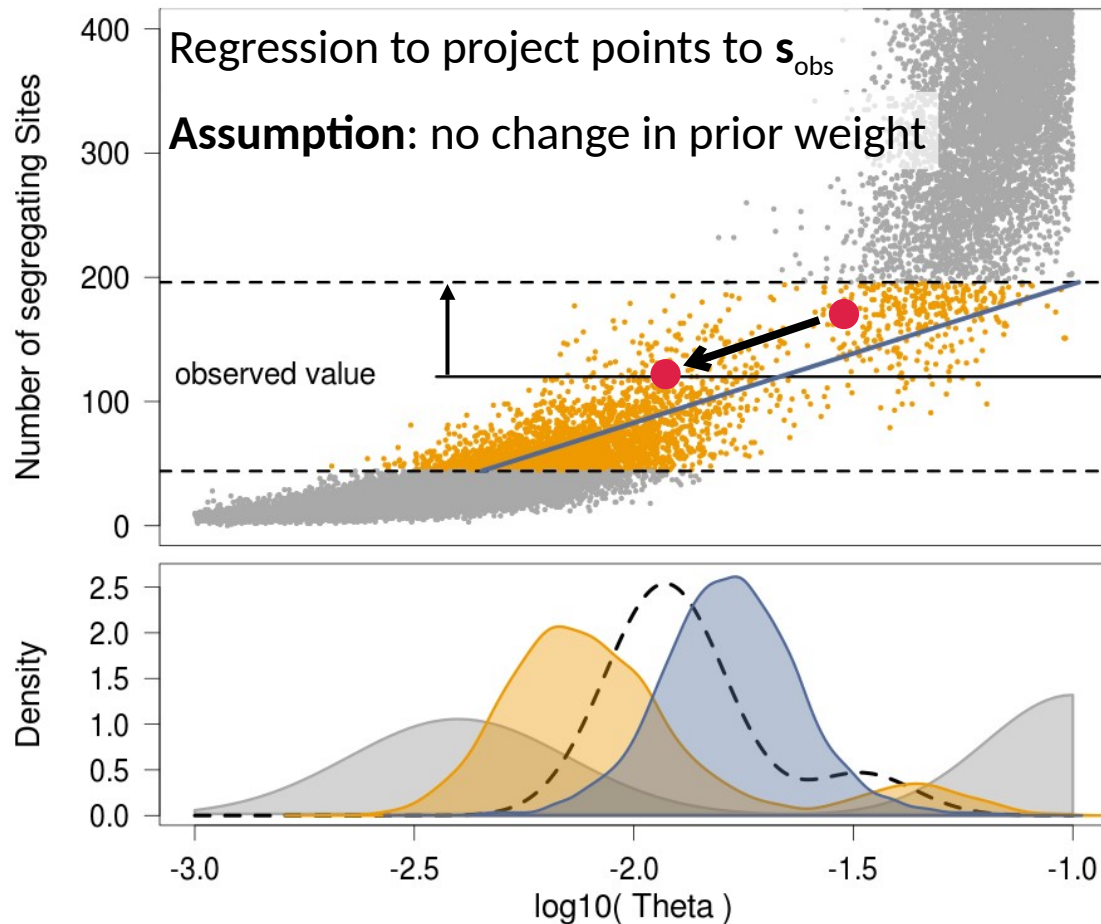
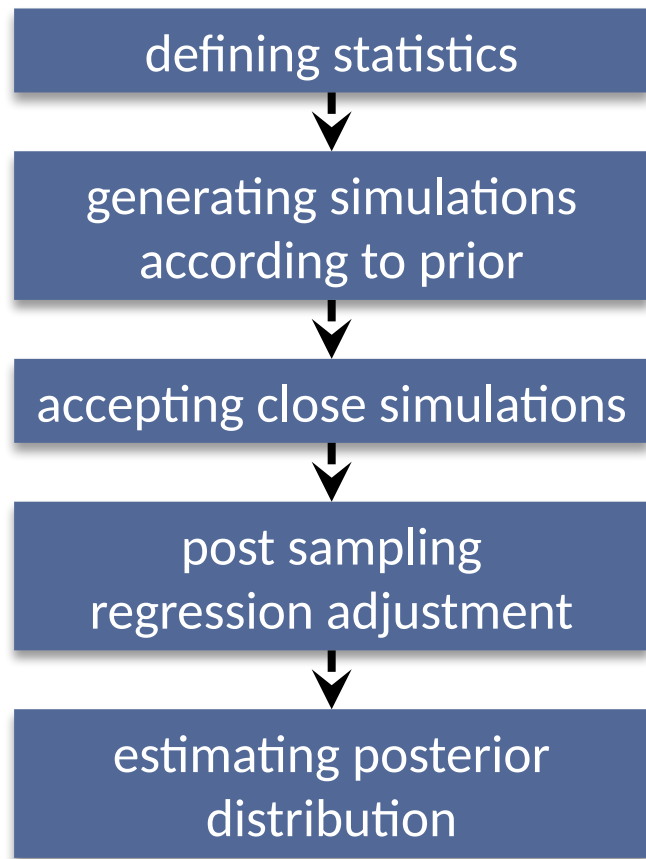
# Approximate Bayesian Computation ABC

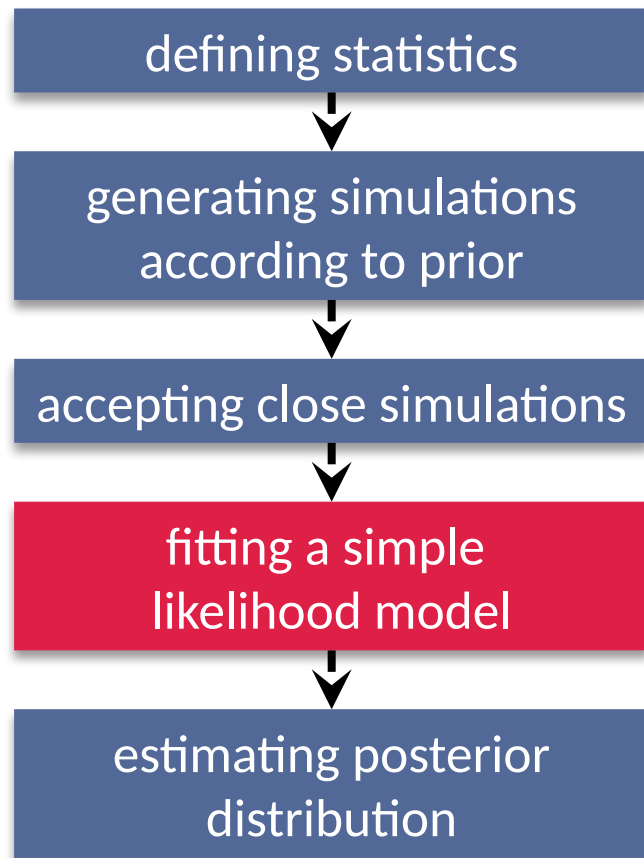


# Approximate Bayesian Computation ABC



# Approximate Bayesian Computation ABC





- It is easy to show that

$$\pi(\boldsymbol{\theta} \mid \mathbf{s}_{\text{obs}}) = f_{\epsilon}(\mathbf{s}_{\text{obs}} \mid \boldsymbol{\theta}) \pi_{\epsilon}(\boldsymbol{\theta})$$

- where  $f_{\epsilon}(\mathbf{s} \mid \boldsymbol{\theta})$  is the truncated likelihood

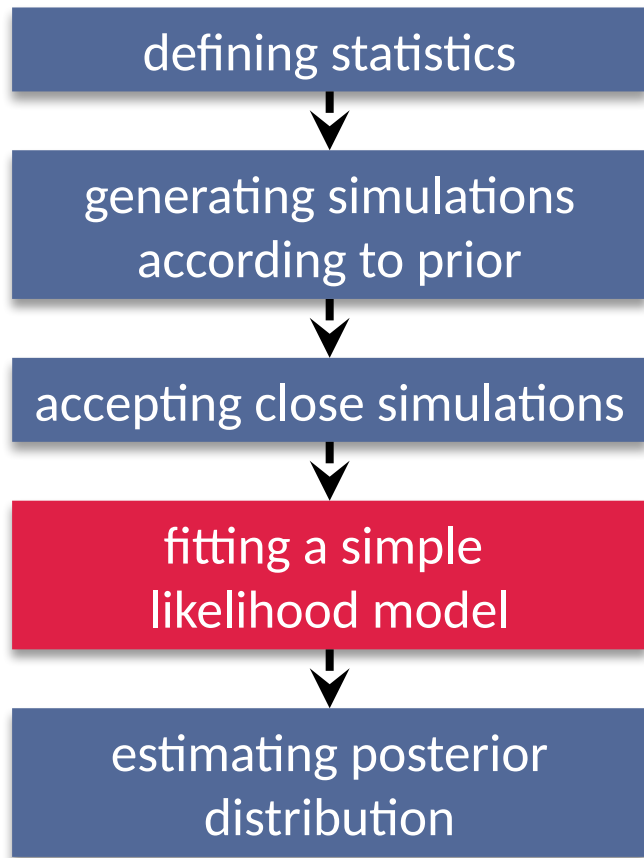
$$f_{\epsilon}(\mathbf{s} \mid \boldsymbol{\theta}) = \underbrace{\text{Ind}(\mathbf{s} \in \mathcal{B}_{\epsilon}(\mathbf{s}_{\text{obs}}))}_{\{\mathbf{s} \in \mathbb{R}^n \mid \text{dist}(\mathbf{s}, \mathbf{s}_{\text{obs}}) < \epsilon\}} \cdot f_{\mathcal{M}}(\mathbf{s} \mid \boldsymbol{\theta})$$

- and  $\pi_{\epsilon}(\boldsymbol{\theta})$  the „truncated prior”

$$\pi_{\epsilon}(\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}) \int_{\mathcal{B}_{\epsilon}} f_{\mathcal{M}}(\mathbf{s} \mid \boldsymbol{\theta}) d\mathbf{s}$$



# ABC-GLM



$$\pi(\boldsymbol{\theta} \mid \mathbf{s}_{\text{obs}}) \propto f_{\epsilon}(\mathbf{s}_{\text{obs}} \mid \boldsymbol{\theta}) \pi_{\epsilon}(\boldsymbol{\theta})$$

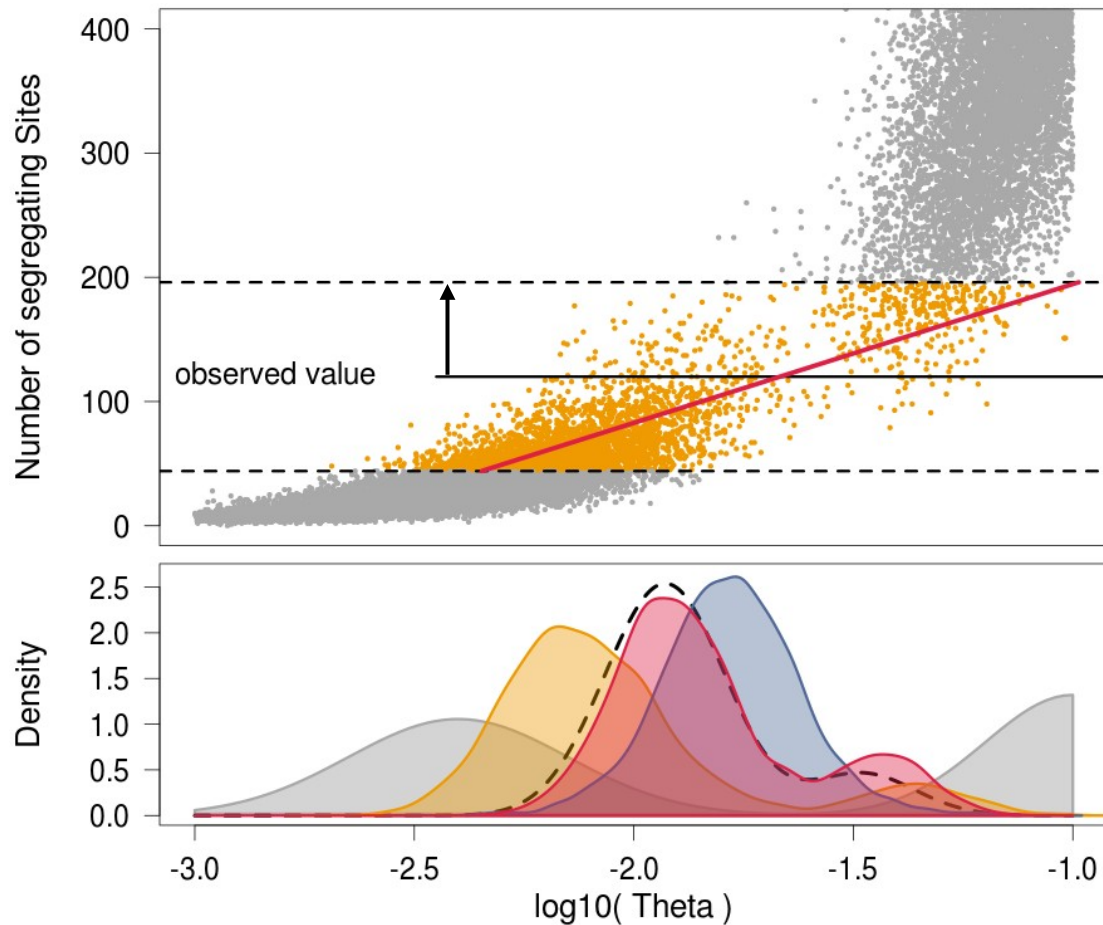
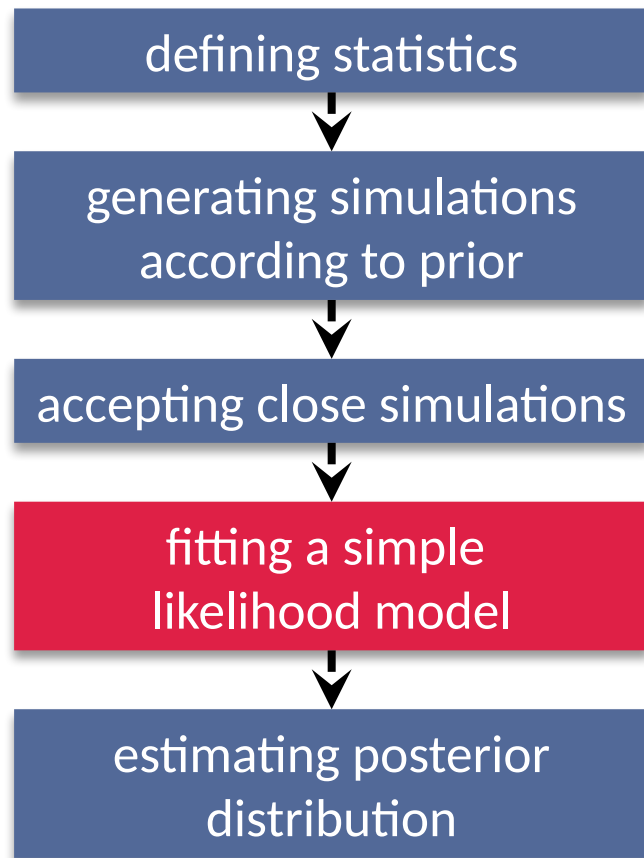
Assume GLM (estimate via OLS)

$$\mathbf{s} \mid \boldsymbol{\theta} = \mathbf{C}\boldsymbol{\theta} + \mathbf{c}_0 + \boldsymbol{\epsilon} \quad \text{with} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_s)$$

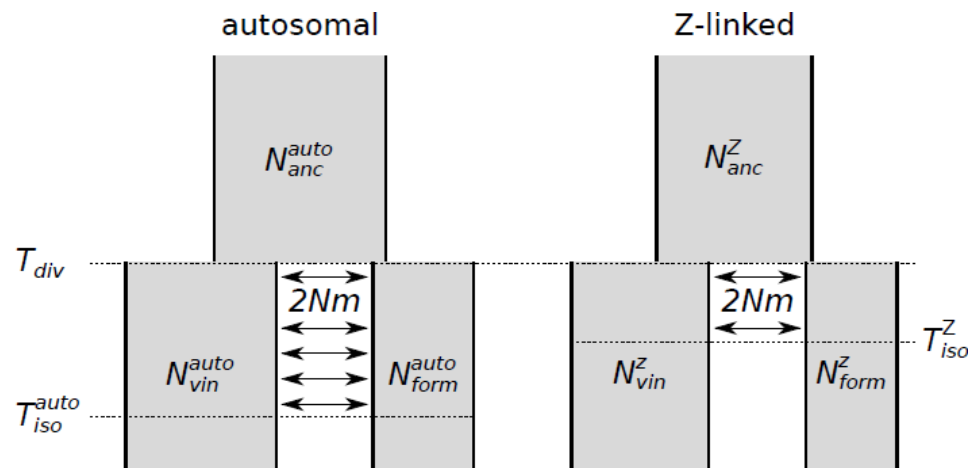
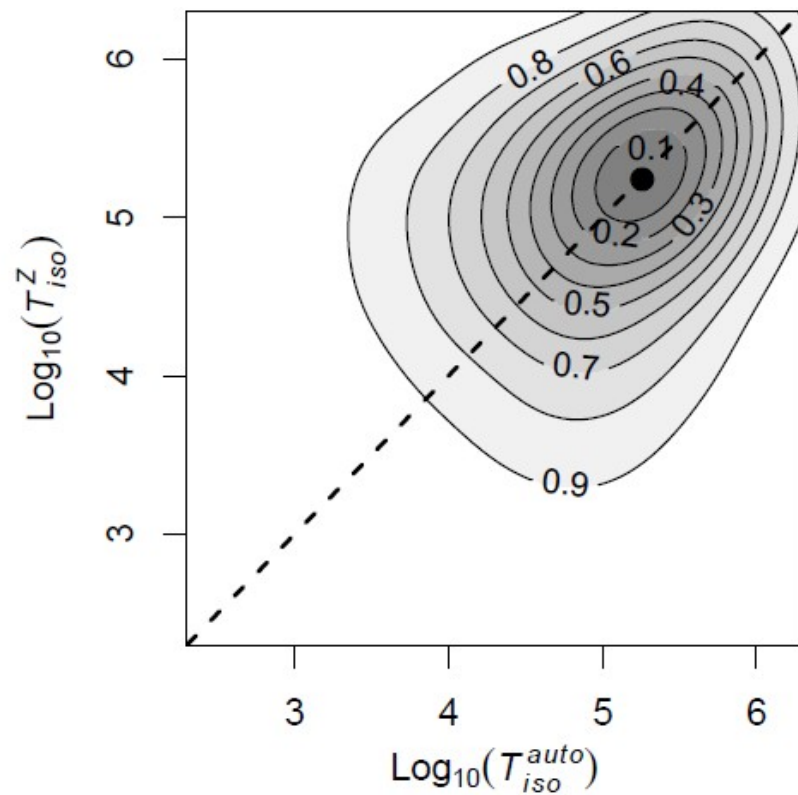
From retained sample using Gaussian peaks

$$\pi_{\epsilon}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{j=1}^N \phi(\boldsymbol{\theta} - \boldsymbol{\theta}^j, \boldsymbol{\Sigma}_{\theta})$$

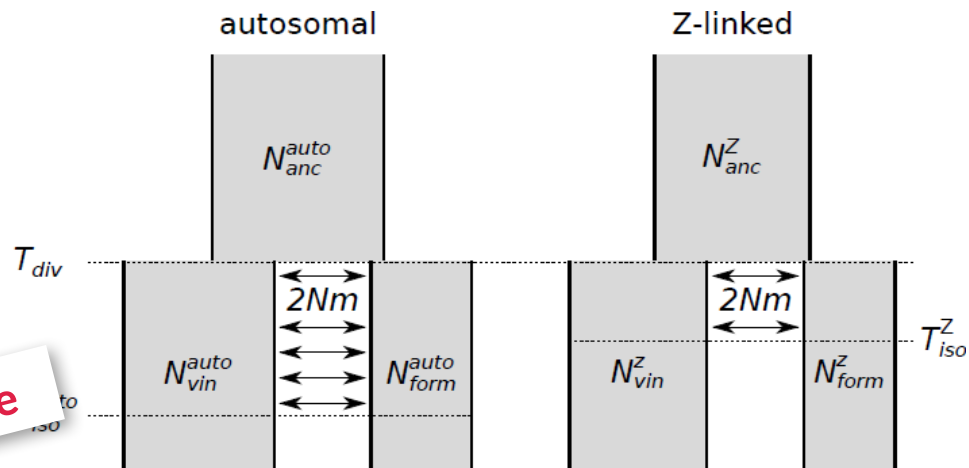
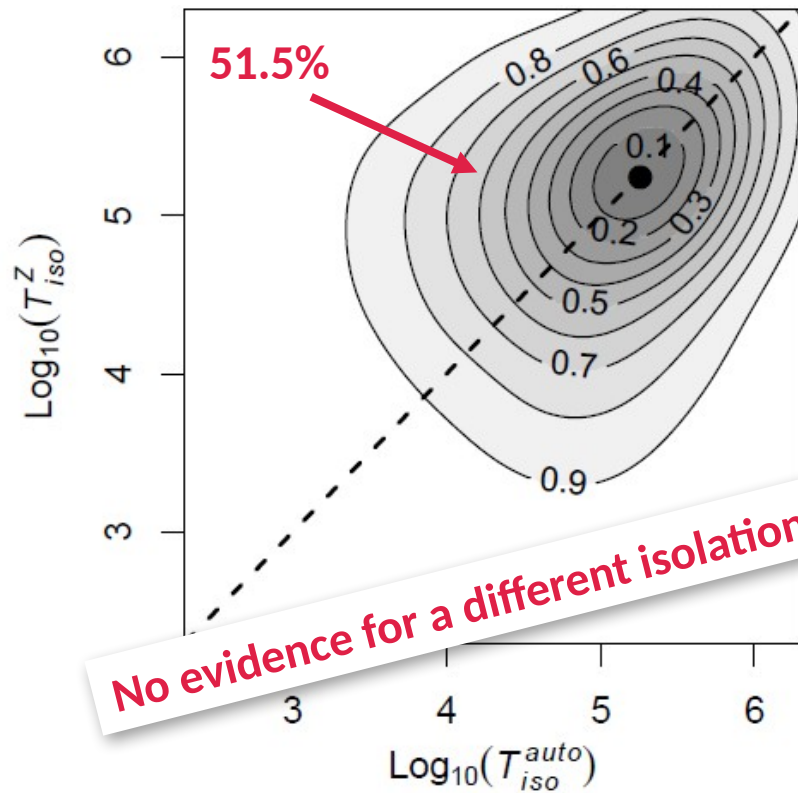
# ABC-GLM



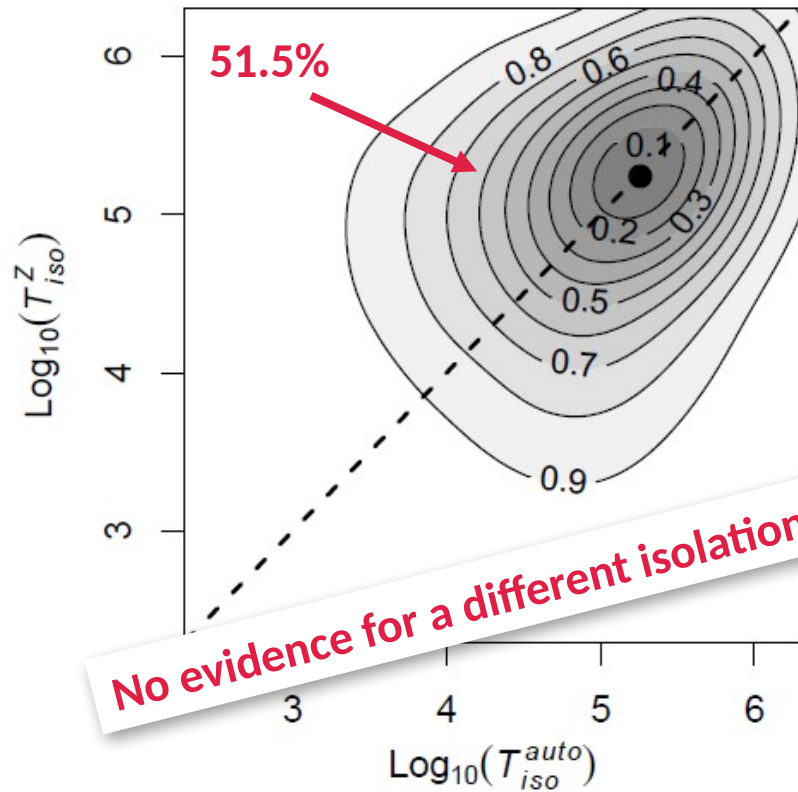
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Joint posterior asymmetry  
observed in simulated data sets

